

Barath R

Electronics & Communication Engineering Student | Embedded Systems & IoT

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PROFESSIONAL SUMMARY

Third-year Electronics and Communication Engineering student with hands-on experience across embedded firmware, bare-metal systems programming, and IoT hardware design. Comfortable working close to the metal -- from register-level RISC-V trap handlers to production-style sensor and communication pipelines -- and building the cloud/dashboard layers that sit on top of them. Seeking an embedded systems internship/role to contribute to real-time and connected hardware products.

EDUCATION

B.E. Electronics and Communication Engineering Rajalakshmi Institute of Technology, Chennai <i>Currently in 3rd year</i>	2024 -- 2028 (Expected) CGPA: 8.65/10
Higher Secondary Certificate (HSC) -- Class XII Montfort Matriculation Higher Secondary School, Chennai <i>Tamil Nadu State Board</i>	2023 -- 2024 Percentage: 95%
Secondary School Leaving Certificate (SSLC) -- Class X Montfort Matriculation Higher Secondary School, Chennai <i>Tamil Nadu State Board</i>	2021 -- 2022 Percentage: 94%

TECHNICAL SKILLS

Hardware & Development Boards: ESP32, ESP32-C6, ESP32-CAM, Raspberry Pi 5, Arduino UNO R3, Altera DE2 Development Kit

PCB Design & EDA Tools: PCB/PCD Design, KiCad, Proteus, Schematic Capture, 2/4-Layer PCB Layout, Routing, DRC/ERC, Gerber & BOM Generation, Soldering & Board Bring-up

Communication Protocols: MQTT, LoRaWAN, I2C, SPI, UART, HTTP/REST, WebSockets

Programming Languages: C, Python, C++, Java, SQL

Cloud & Dashboards: HiveMQ, ThingSpeak, Firebase, BlynkIoT, Google Cloud

Tools: Git, VS Code, PlatformIO, ESP-IDF (v6), Arduino IDE, MATLAB, Scilab, CST Studio, Keysight ADS

Android/OS: ADB, custom OS/ROM flashing

FEATURED PROJECTS

Bare-Metal RISC-V Trap Handler & Context Switcher <i>Tech Stack: RISC-V Assembly, direct MMIO, no RTOS</i>	ESP32-C6
- Implemented a trap handler and timer-based context switcher entirely in assembly with no FreeRTOS or vendor libraries - Worked directly with memory-mapped I/O registers for interrupt handling and task switching	
Dual-Axis Solar Tracker <i>Tech Stack: Servo Motors, LDR Sensors, Microcontroller Control</i>	Embedded System
- Built a dual-axis solar panel tracking system to maximize sunlight incidence using closed-loop sensor feedback - Implemented motor control logic for continuous azimuth and elevation tracking	
Digital Twin for Industrial Monitoring <i>Tech Stack: Sensor Fusion, 5G Connectivity, Real-Time Dashboard</i>	5G-Integrated IoT System
- Developing a digital twin platform for industrial equipment monitoring with 5G-integrated data pipelines - Designing real-time sensor fusion and visualization for predictive maintenance use cases	
Conformal Helmet-Integrated Antenna Array <i>Tech Stack: Microstrip Patch Array, Dual-Band UHF/L-Band</i>	RF / Antenna Design
- Designing a flexible microstrip patch antenna array conformal to a helmet surface, replacing bulky whip antennas - Targeting dual-band UHF and L-band support with an RF shielding layer and ruggedized coax interface to existing radio/camera systems	

TECHNICAL EXPERIENCE

- Conceptualized, developed, and iterated on multiple embedded and IoT systems end-to-end, from hardware bring-up to firmware and dashboards
- Active organizer with IETE Students' Forum (ISF), coordinating technical hackathons and events

LANGUAGES

English (Fluent) • Tamil (Native)